

Reproduction Instructions

This document provides step-by-step instructions for reproducing the KoeiNet workflow using the Nanluoguxiang use case, from the original input data to the generated analytical outputs. These outputs correspond directly to the results, metrics, and figures reported in the relevant sections of the manuscript.

PART ONE: Prepare the Project, Environment, and Data

1. Obtain the Project Code and Navigate to the Project Directory

1.1 Option A: Clone the Repository Using Git

```
git clone https://github.com/mark6880/KoeiNet.git
cd KoeiNet
```

1.2 Option B: Download and Extract the ZIP Archive

Download the ZIP archive from the repository page, extract it, and navigate to the root directory of the extracted KoeiNet project.

2. Create and Activate a Virtual Environment

2.1 Create a Virtual Environment

It is recommended to use Python's built-in `venv` module to create an independent and isolated environment:

```
python -m venv venv
```

2.2 Activate the Virtual Environment

Windows (PowerShell / CMD):

```
.\venv\Scripts\activate
```

macOS / Linux:

```
source venv/bin/activate
```

3. Install the Required Python Packages

Optionally upgrade pip within the virtual environment:

```
python -m pip install --upgrade pip
```

Install all dependencies listed in requirements.txt:

```
pip install -r requirements.txt
```

4. Install FFmpeg for Video and Audio Processing

FFmpeg is required for video frame extraction and audio processing. Select the appropriate installation method for your operating system.

- ✓ Windows: Download and extract FFmpeg, and then either place ffmpeg.exe in the repository root directory (KoeiNet/) or add it to the system PATH:

<https://www.gyan.dev/ffmpeg/builds/>

- ✓ macOS: Install FFmpeg using Homebrew:

```
brew install ffmpeg
```

- ✓ Ubuntu/Debian: Install FFmpeg using the apt package manager:

```
sudo apt update && sudo apt install -y ffmpeg
```

5. Download and Organize the Raw Data

The original panoramic video and corresponding GPS trajectory data used in the Nanluoguxiang use case are available through the following Figshare link:

<https://figshare.com/s/1c882d0430802b474d8c>

After downloading the data, create an input/ directory under the KoeiNet project root and place

both files in this directory.

Academic-use statement: These data files are provided exclusively for academic research and must not be used for commercial purposes or any other unauthorized purposes.

PART TWO: Run the Code and Generate Continuous Multisensory

Street Assessment Outputs

- This part covers the core functionality of KoeiNet and reproduces the visual, emotional, and soundscape assessment results and visualizations for the Nanluoguxiang use case.
- Follow the steps below to reproduce the metrics and Figures 5–8 reported in Section 4 of the manuscript from the original input data.

1. Run Program 01

After confirming that the original video has been placed in the input/ directory, run Program 01:

```
python scripts/run_program_01.py
```

Once the input video has been successfully detected, the terminal will display a processing-start message and real-time progress information.

2. Locate the Output Directory

The default output directory is:

```
output/<VIDEO_NAME>/
```

After processing is completed, multiple subdirectories will be generated under this output directory.

3. Review the Generated Results

Review the integrated outputs generated by KoeiNet, including result data files, figures, and interactive visualizations. The quantitative metrics reported in Section 4.2 are derived directly from the generated CSV files through straightforward arithmetic operations, such as calculating means, minima, maxima, standard deviations, coefficients of variation, proportions, and

occurrence counts.

(1) Review the Visual and Emotional Assessment Outputs

The results are written to the stats/ directory, with the following structure:

```
output/<VIDEO_NAME>/stats/  
├─ color_analysis/  
├─ emotion/  
├─ green_view/  
├─ people_count/  
└─ visual_elements/
```

Using green_view/ as an example, the directory structure is:

```
output/<VIDEO_NAME>/stats/green_view/  
├─ green_view_index.csv  
├─ green_view_index_chart.png  
└─ green_view_index_interactive.html
```

Correspondence to the Nanluoguxiang Use Case: The CSV files contain the complete assessment data used for the Nanluoguxiang use case. The PNG files visualize these results, while the HTML files provide interactive visualizations. Review the individual observations directly from the CSV files and calculate the reported summary statistics from these values.

● Reproduce the Manuscript Metrics and Figures

Section 4.2.1 Visual Assessment:

- ✓ Use the outputs in output/<VIDEO_NAME>/stats/visual_elements/, output/<VIDEO_NAME>/stats/color_analysis/, and output/<VIDEO_NAME>/stats/green_view/.
- ✓ Calculate the mean proportions of the visual-element categories and area-weighted color categories from the corresponding CSV files. For GVI, calculate the mean, minimum, maximum, standard deviation, and coefficient of variation from the continuous GVI values, with the coefficient of variation calculated as the standard deviation divided by the mean.

- ✓ Figure 5 in the manuscript was assembled from the figures generated in these output directories; Figure 6 in the manuscript was assembled from the figures generated in the corresponding output directory and the relevant frames from the input video.

Section 4.2.2 Emotional Assessment:

- ✓ Use the outputs in output/<VIDEO_NAME>/stats/emotion/.
- ✓ Calculate the mean and standard deviation of each emotional dimension from the continuous values recorded in the corresponding CSV files. Values for individual locations or frames discussed in the manuscript can be obtained directly from the corresponding records.
- ✓ Figure 7 in the manuscript was assembled from the figures generated in this output directory and the relevant frames from the input video.

(2) Review the Soundscape Assessment Outputs

The results are written to the audio_events/ directory, with the following structure:

```
output/<VIDEO_NAME>/audio_events/  
├─ <VIDEO_NAME>.wav  
├─ audio_events_proportion.csv  
├─ ...  
└─ top_events.png
```

Correspondence to the Nanluoguxiang Use Case: The CSV file contains the complete result data for the Nanluoguxiang use case, while the PNG file provides a visualization based on these results.

● Reproduce the Manuscript Metrics and Figures

Section 4.2.3 Soundscape Assessment:

- ✓ Use the outputs in output/<VIDEO_NAME>/audio_events/.
- ✓ Calculate the mean probabilities of the grouped soundscape categories and individual sound events across segments from the corresponding CSV files, and obtain event occurrence counts by counting the segments in which each event is recorded.
- ✓ The figure generated in this output directory was used directly as Figure 8 in the manuscript.

PART THREE: Run the Code and Generate Problem-Episode

Identification Outputs (Optional)

- This part presents a derived application of KoeiNet. Problem episodes are identified by combining the continuous multisensory street assessment results with human-adjudicated annotations. The repository provides the annotation CSV file used in the Nanluoguxiang use case, together with a tutorial for creating custom annotations in the Appendix.
- Follow the steps below to reproduce the problem-episode identification results reported in Section 5 and detailed in Supplementary Section C of the manuscript.

1. Verify the Required Inputs

Confirm that the required Program 01 output directory is available at:

```
output/<VIDEO_NAME>/
```

Confirm that the required human-adjudicated annotation CSV file is available at:

```
usecase/validation/final_annotation_labels_adjudicated.csv
```

2. Run Program 02

After confirming that both the Program 01 output directory and the human-adjudicated annotation CSV file exist, run the following command from the project root directory:

```
python scripts/run_program_02.py --program_01_output "output/<VIDEO_NAME>" --  
annotation_csv "usecase/validation/final_annotation_labels_adjudicated.csv"
```

3. Locate and Review the Problem-Detection Outputs

By default, the results are written to:

```
output/<VIDEO_NAME>/problem_detection/
```

The main output files include:

```
output/<VIDEO_NAME>/problem_detection/
```

```
├─ segment_problem_priority.csv
```

```
├─ problem_episodes.csv
```

```
├─ ...
```

```
└─ problem_detection_summary.md
```

Correspondence to the Nanluoguxiang Use Case: These outputs reproduce the segment-level priority assessment and continuous problem-episode identification reported in Section 5 and Supplementary Section C. The segment-level priority results are recorded in `segment_problem_priority.csv`, while the identified continuous problem episodes are recorded in `problem_episodes.csv`. Together, these outputs provide the results underlying the problem-episode patterns presented in Figure S3.

Appendix: Generate an Annotation Template, Complete Manual

Annotations, and Run Problem Detection Through the GUI

1. Install the Required GUI Dependencies

Install all additional dependencies listed in `requirements_gui.txt`:

```
pip install -r requirements_gui.txt
```

2. Generate an Annotation Template

```
python scripts/run_program_02.py --program_01_output "<PROGRAM01_OUTPUT_DIR>" -  
-create_annotation_template
```

Example:

```
python scripts/run_program_02.py --program_01_output "output/<VIDEO_NAME>" --create_annotation_template
```

This command generates an annotation-template CSV file under the validation/ directory within the Program 01 output directory.

3. Launch the Program 02 GUI

```
python scripts/run_program_02.py --launch_gui
```

4. Complete GUI Workflow

The following steps must be completed in sequence:

- 1) Select the Program 01 output directory, for example, output/<VIDEO_NAME>/.
- 2) When no annotation file is available, click Create Annotation File. Alternatively, the template generated in Step 2 will appear automatically.
- 3) On the Annotation page, enter or verify the human-scoring fields for each record. Common fields include:
 - a) comfort_score (1–5)
 - b) vitality_score (1–5)
 - c) soundscape_pleasantness (1–5)
 - d) soundscape_eventfulness (1–5)
 - e) overall_problem_severity (1–5)
 - f) confidence_score (1–5)
 - g) main_problem_labels (semicolon-separated)
 - h) annotator_notes
- 4) After completing the annotations, click Save Annotation. The file will be saved under output/<VIDEO_NAME>/validation/.
- 5) Navigate to the Coefficients page, load or retain the default configs/street_type_coefficients.yaml file, adjust the coefficients as needed, and then save or apply the settings.
- 6) Set the detection parameters in the GUI parameter panel: Top K, Priority Threshold, and Max Gap Seconds. Example settings are Top K = 2, Priority Threshold = 0.4, and Max Gap Seconds = 5.
- 7) Click Run Problem Detection, wait for processing to complete, and review the logs and progress information.